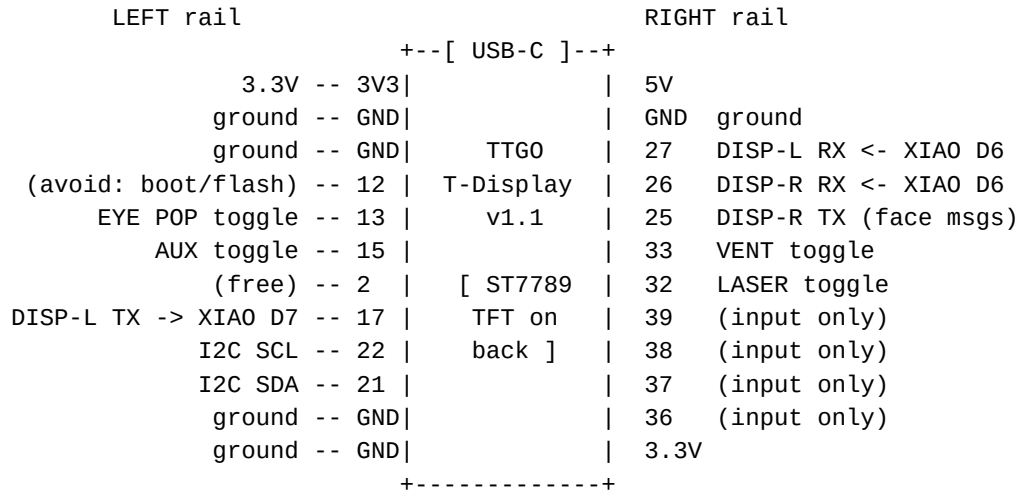


# j4\_controller -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)



on-board (no header pin): GPIO34 = battery sense, BOOT = GPIO00,  
GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23  
I2C bus: ADS\_01 @ 0x48, ADS\_02 @ 0x49, ADS\_03 @ 0x4A, ADS\_04 @ 0x4B,  
PCF8574 keypad\_left @ 0x21, keypad\_right @ 0x20  
toggles: INPUT\_PULLUP, switch closes to GND (ON = LOW)

| GPIO | Function                                                            |
|------|---------------------------------------------------------------------|
| 21   | I2C SDA (ADS1115 x4, PCF8574 keypads x2)                            |
| 22   | I2C SCL                                                             |
| 17   | Serial1 TX to j4_display_left (XIAO D7)                             |
| 27   | Serial1 RX from j4_display_left (XIAO D6)                           |
| 25   | Serial2 TX to j4_display_right (XIAO D7, face-preset messages)      |
| 26   | Serial2 RX from j4_display_right (XIAO D6, pot feed)                |
| 32   | LASER toggle (INPUT_PULLUP, switch closes to GND)                   |
| 33   | VENT toggle (INPUT_PULLUP, switch closes to GND)                    |
| 13   | EYE POP toggle (INPUT_PULLUP, switch closes to GND)                 |
| 15   | AUX toggle, next to the right joystick (INPUT_PULLUP, unassigned)   |
| 34   | Battery voltage sense (ADC input, read-only)                        |
| 35   | Screen-cycle button (TTGO built-in; data / MAC / connection status) |

# j4\_controller -- ADS1115 pin diagrams

pot wiper to the A-pin; outer legs to 3.3V and GND

ADS\_01 @ 0x48 (ADDR -> GND)

both joysticks

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|      |                             |                                   |
|------|-----------------------------|-----------------------------------|
| VDD  | 3.3V                        |                                   |
| GND  | GND                         |                                   |
| SCL  | TTGO GPIO 22 (I2C SCL)      |                                   |
| SDA  | TTGO GPIO 21 (I2C SDA)      |                                   |
| ADDR | GND = address 0x48          |                                   |
| ALRT | not connected               |                                   |
| A0   | neck joystick X wiper       | -> neck L/R differential mix      |
| A1   | neck joystick Y (jaw) wiper | -> neck L/R base height           |
| A2   | eyes joystick X wiper       | -> eyes pan servo (PCA9685 ch 3)  |
| A3   | eyes joystick Y wiper       | -> eyes tilt servo (PCA9685 ch 4) |

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ADS\_02 @ 0x49 (ADDR -> VDD)

four face pots, left bank

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|      |                            |                 |
|------|----------------------------|-----------------|
| VDD  | 3.3V                       |                 |
| GND  | GND                        |                 |
| SCL  | TTGO GPIO 22 (I2C SCL)     |                 |
| SDA  | TTGO GPIO 21 (I2C SDA)     |                 |
| ADDR | VDD = address 0x49         |                 |
| ALRT | not connected              |                 |
| A0   | Eyebrow L pot wiper        | -> PCA9685 ch 6 |
| A1   | Eyebrow R pot wiper        | -> PCA9685 ch 7 |
| A2   | Basket Eyebrow L pot wiper | -> PCA9685 ch 8 |
| A3   | Basket Eyebrow R pot wiper | -> PCA9685 ch 9 |

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ADS\_03 @ 0x4A (ADDR -> SDA)

three face pots, right bank

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|      |                                                            |                  |
|------|------------------------------------------------------------|------------------|
| VDD  | 3.3V                                                       |                  |
| GND  | GND                                                        |                  |
| SCL  | TTGO GPIO 22 (I2C SCL)                                     |                  |
| SDA  | TTGO GPIO 21 (I2C SDA)                                     |                  |
| ADDR | SDA = address 0x4A                                         |                  |
| ALRT | not connected                                              |                  |
| A0   | Nose pot wiper (up/down)                                   | -> PCA9685 ch 10 |
| A1   | FAULTY channel, nothing connected, left unread in firmware |                  |
| A2   | Bottom Eyelid L pot wiper                                  | -> PCA9685 ch 12 |
| A3   | Bottom Eyelid R pot wiper                                  | -> PCA9685 ch 13 |

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ADS\_04 @ 0x4B (ADDR -> SCL)

neck-pivot + faders + Nose Basket

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|      |                        |                                           |
|------|------------------------|-------------------------------------------|
| VDD  | 3.3V                   |                                           |
| GND  | GND                    |                                           |
| SCL  | TTGO GPIO 22 (I2C SCL) |                                           |
| SDA  | TTGO GPIO 21 (I2C SDA) |                                           |
| ADDR | SCL = address 0x4B     |                                           |
| ALRT | not connected          |                                           |
| A0   | neck-pivot pot wiper   | -> nP on the stepper link (0-3200)        |
| A1   | fader_left wiper       | -> neck pivot (doubles ADS_04 A0, 0-3200) |
| A2   | fader_right wiper      | -> iris (doubles j4_display_right IRIS)   |
| A3   | Nose Basket pot wiper  | -> PCA9685 ch 11 (single source)          |

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